

UniVision Project Progress Report

June 2026

1 Introduction

Over the first half of the SoC, the UniVision project evolved from a basic frontend application into the foundation of a computer vision pipeline. The work included developing a React interface, building REST APIs with FastAPI, designing a visual workflow editor, and learning image processing using OpenCV and NumPy. Each stage built upon the previous one, resulting in a modular and extensible system.

2 Frontend Development

The frontend was developed using React and TypeScript with a component-based architecture. Reusable components improved code organization, while state management and props enabled dynamic interaction between different parts of the application. React Router was used for seamless navigation between pages without reloading the application.

This phase also involved debugging common React issues such as routing errors, dependency conflicts, and component rendering problems.

3 Backend and Workflow Design

Backend development was carried out using FastAPI, with API request and response formats defined before implementation. Pydantic models were used for input validation, ensuring invalid requests were rejected automatically, while appropriate HTTP status codes handled different response scenarios.

The project also introduced a visual workflow editor using React Flow, where image processing tasks were represented as connected nodes. Workflow execution was validated using Directed Acyclic Graphs (DAGs), and Kahn's Algorithm determined the correct execution order while detecting cyclic dependencies. React state enabled users to create and modify workflows dynamically.

4 Image Processing Fundamentals

The final phase focused on understanding images as multidimensional arrays using OpenCV and NumPy. Basic preprocessing operations were implemented to prepare images for later computer vision tasks.

- Loading and displaying images
- Inspecting image dimensions and channels
- Resizing images
- Cropping regions of interest
- Converting images to grayscale
- Normalizing pixel values

These operations provide the basic preprocessing pipeline required before applying computer vision or machine learning algorithms.

5 Challenges Faced

The project involved several debugging challenges. React applications occasionally failed due to routing errors, duplicate package installations, and CSS layout issues affecting the React Flow workspace. During image processing, OpenCV was initially unavailable because the Python virtual environment had not been activated. Resolving these issues improved understanding of dependency management and development workflows.

6 Conclusion

The work completed during the first half of the SoC established the core framework of UniVision. The project now consists of a React frontend, validated REST APIs, a workflow editor for organizing processing steps, and an image preprocessing module. This foundation is ready for integrating computer vision models in the upcoming weeks.